

Multiple Choice

1. **Answer:** B

Options: A) Yes, with $p=2$.; B) Yes, with $p=3$.; C) Yes, with $p=5$.; D) No.

Note: Correct option: B - Yes, with $p=3$.

2. **Answer:** D

Options: A) True, True; B) False, False; C) True, False; D) False, True

Note: Correct option: D - False, True

3. **Answer:** B

Options: A) If A is true, then B is false.; B) If A is false, then B is false.; C) If A is false, then B is true.; D) Both A and B are true.

Note: Correct option: B - If A is false, then B is false.

4. **Answer:** B

Options: A) 24; B) 30; C) 32; D) 36

Note: Correct option: B - 30

5. **Answer:** A

Options: A) 0; B) 3; C) 12; D) 30

Note: Correct option: A - 0

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '-1'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $e^2/2$ '.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'a point'.

Long Form Response

11. **Answer:** The remainder when $37+49+14$ is divided by 12 is the same as the remainder when the congruences of each number mod 12 are summed and then divided by 12. In other words, since we have $37 \equiv 1 \pmod{12}$
 $49 \equiv 1 \pmod{12}$
 $14 \equiv 2 \pmod{12}$ then therefore, $37 + 49 + 14 \equiv 1 + 1 + 2 \equiv 4 \pmod{12}$.
Note: Students should show all steps in their mathematical solution.
12. **Answer:** 35. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.